Customer Shopping Behavior Analysis Report

1. Project Overview

In this project, I analyzed customer shopping behavior using a retail dataset containing around 3,900 transactions across multiple product categories. The aim was to understand how different factors such as demographics, discounts, and subscription status influence purchasing patterns and overall revenue.

The project follows a complete data analysis workflow, starting from data cleaning in Python to business analysis in SQL and visualization using Power BI.

2. Dataset Summary

- Total Records: 3,900
- Total Columns: 18

Key Features:

- **Customer Details:** Age, Gender, Location, Subscription Status
- **Purchase Information:** Item Purchased, Category, Purchase Amount
- **Behavioral Data:** Discount Applied, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

Data Quality:

- Found missing values in review_rating column (37 entries), which were handled during preprocessing

3. Data Cleaning & Preparation (Python)

The dataset was first explored and cleaned using Python:

- Loaded data using pandas and checked structure with .info() and .describe()
- Identified missing values and handled them using **median imputation based on product category**
- Renamed columns into **snake_case format** for better readability

- Performed feature engineering:
    - Created age_group to segment customers by age
    - Converted purchase frequency into numeric values (purchase_frequency_days)
    - Added spending_level to classify customers based on purchase amount
    - Created is_repeat_customer to identify frequent buyers
- Checked for redundant columns and removed promo_code_used after validation
- Loaded cleaned data into PostgreSQL for further analysis

4. Data Analysis (SQL)

SQL was used to answer business-related questions:

- Analyzed total revenue generated by different genders
- Identified customers who used discounts but still spent above average
- Found top-rated products based on customer reviews
- Compared purchase behavior across different shipping methods
- Evaluated differences between subscribed and non-subscribed customers
- Identified products with high dependency on discounts
- Segmented customers into New, Returning, and Loyal groups
- Ranked top-performing products within each category
- Analyzed relationship between repeat purchases and subscription status
- Calculated revenue contribution across different age groups

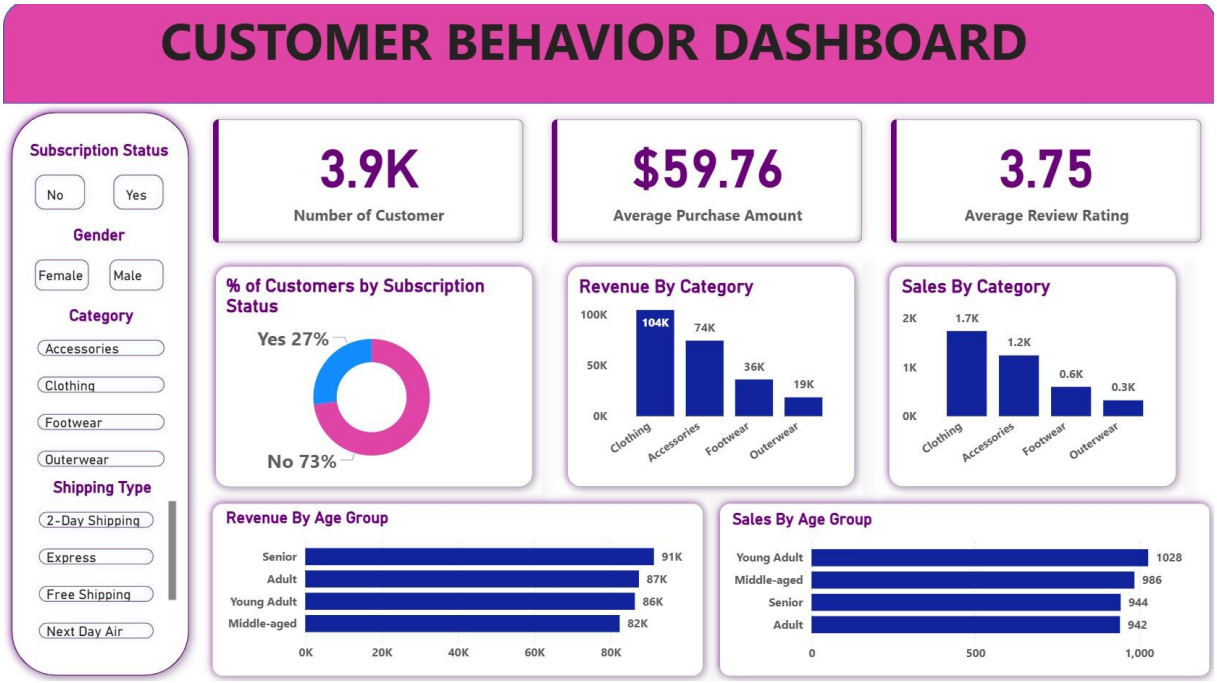
5. Dashboard (Power BI)

An interactive dashboard was created in Power BI to present the insights visually.

The dashboard includes:

- Revenue breakdown
- Customer segmentation
- Product performance analysis

Filters were added to allow users to explore data by category, age group, and subscription status.



6. Key Insights

- Customers who used discounts often spent amounts equal to or above average
- Loyal customers contributed a major portion of total revenue
- Some product categories showed higher repeat purchase behavior
- Subscription status influenced customer spending patterns

7. Business Recommendations

- Improve customer retention by targeting loyal customers with rewards
- Optimize discount strategies to maintain profitability
- Focus marketing efforts on high-performing products
- Enhance subscription benefits to increase long-term engagement

8. Conclusion

This project helped me understand how to work with structured data and apply different tools like Python, SQL, and Power BI to extract meaningful insights. It also improved my ability to think from a business perspective while analyzing data.